Data Mining Assignment 3: Clustering

Dataset Selection

Three real datasets were selected and standardized before clustering. The selection hypotheses are density-based clustering for Breast Cancer, hierarchical clustering for Iris, and prototype-based clustering for Wine.

dataset	name	samples	features	classes	source
breast_cancer	Breast Cancer Wisconsin Diagnostic	569	30	2	https://archive.ics.uci.edu/datas
iris	Iris	150	4	3	https://archive.ics.uci.edu/datas
wine	Wine Recognition	178	13	3	https://archive.ics.uci.edu/datas

Theory Notes

DBSCAN advantages: it discovers arbitrary-shaped clusters and identifies noise without pre-setting the number of clusters. DBSCAN disadvantages: it is sensitive to eps/minPts and struggles with varying density. OPTICS was selected as the density-based alternative because it orders points across multiple density levels and is less dependent on a single global eps.

Hierarchical clustering categories are agglomerative and divisive. Agglomerative methods are used more widely because they are simpler to implement, easier to visualize with dendrograms, and computationally more practical for ordinary tabular datasets.

KMeans limitations: outliers can pull centroids away from dense groups, and distance concentration harms performance as dimensionality increases. KMedoids mitigates outliers by choosing observed representative points. PCA+KMeans mitigates dimensionality by clustering after variance-preserving projection.

Best Empirical Results

Density: OPTICS

dataset	algorithm	clusters_found	silhouette	davies_bouldin	nmi	adjusted_rand	purity
breast_cancer	OPTICS	1	NaN	NaN	0.080204	0.046674	0.659051
iris	OPTICS	2	0.820590	0.250707	0.207162	0.033727	0.480000
wine	OPTICS	2	0.539483	0.712177	0.346856	0.172411	0.634831

Hierarchical: Agglomerative Linkages

dataset	algorithm	clusters_found	silhouette	davies_bouldin	nmi	adjusted_rand	puri
breast_cancer	single	2	0.660667	0.449709	0.010182	0.004828	0.6309
iris	single	3	0.504646	0.492925	0.720118	0.558371	0.6666
wine	minimum_variance	3	0.277444	1.418592	0.786465	0.789933	0.9269

Prototype-Based Algorithms

dataset	algorithm	clusters_found	silhouette	davies_bouldin	nmi	adjusted_rand	purity
breast_cancer	KMedoids	2	0.349134	1.324375	0.491071	0.607897	0.891037
iris	KMedoids	2	0.581750	0.593313	0.733680	0.568116	0.666667
wine	KMeans_baseline	3	0.284859	1.389188	0.875894	0.897495	0.966292

Random Data Validation

Observed cluster silhouettes were compared with silhouettes from random Gaussian data having the same feature means and standard deviations. Low empirical p-values indicate clustering structure stronger than random baseline.

dataset	family	observed_silhouette	random_mean_silhouette	random_repetitions	em
breast_cancer	density	NaN	NaN	0	
iris	density	0.820590	0.528617	1	
wine	density	0.539483	NaN	0	
breast_cancer	hierarchical	0.660667	0.015785	5	
iris	hierarchical	0.504646	0.152749	5	
wine	hierarchical	0.277444	0.052290	5	
breast_cancer	prototype	0.349134	0.030738	5	
iris	prototype	0.581750	0.192228	5	
wine	prototype	0.284859	0.068910	5	

Visualizations

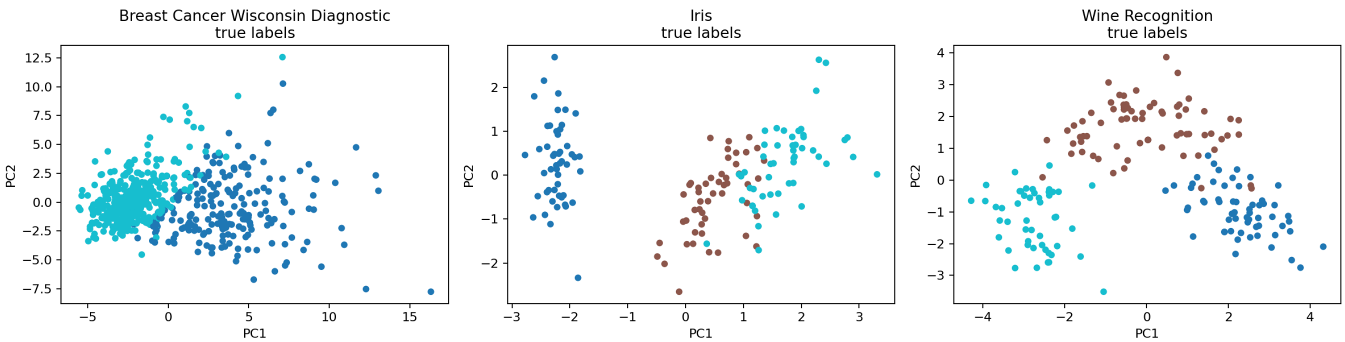

Figure 1: PCA overview with true labels.

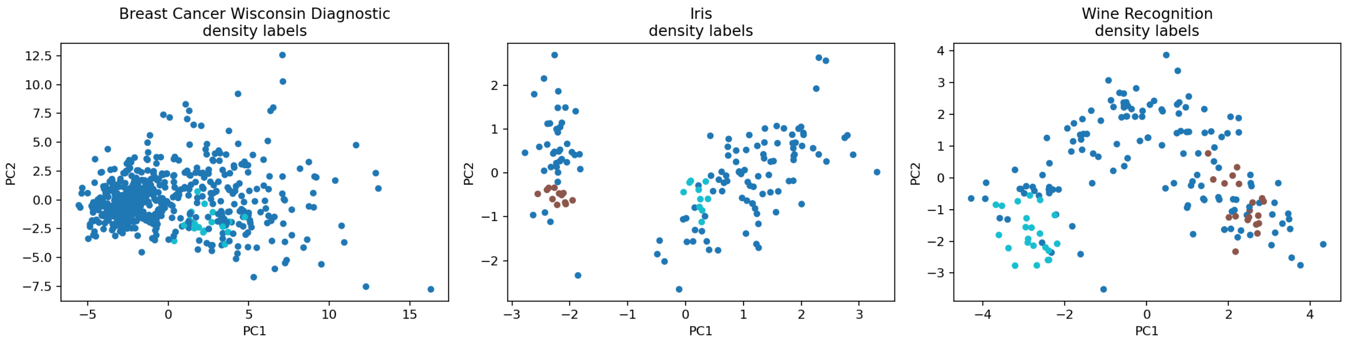

Figure 2: Best OPTICS clusters.

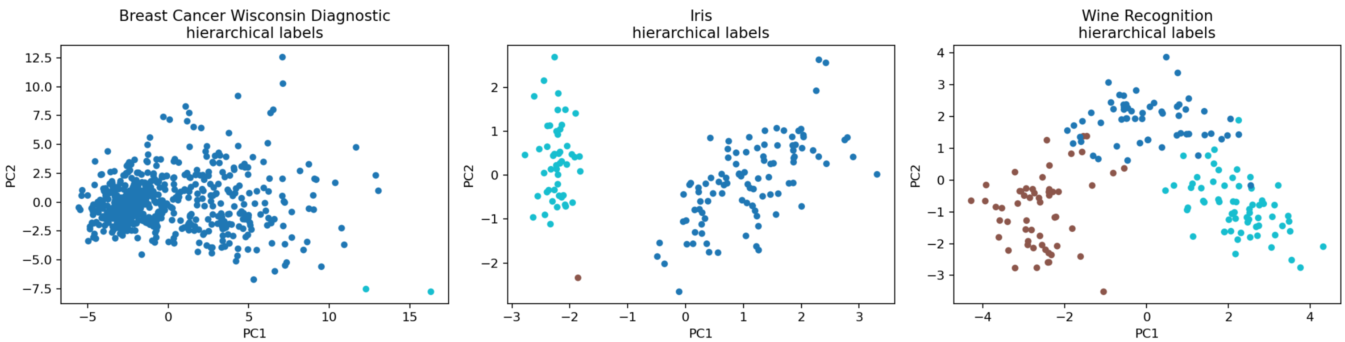

Figure 3: Best hierarchical clusters.

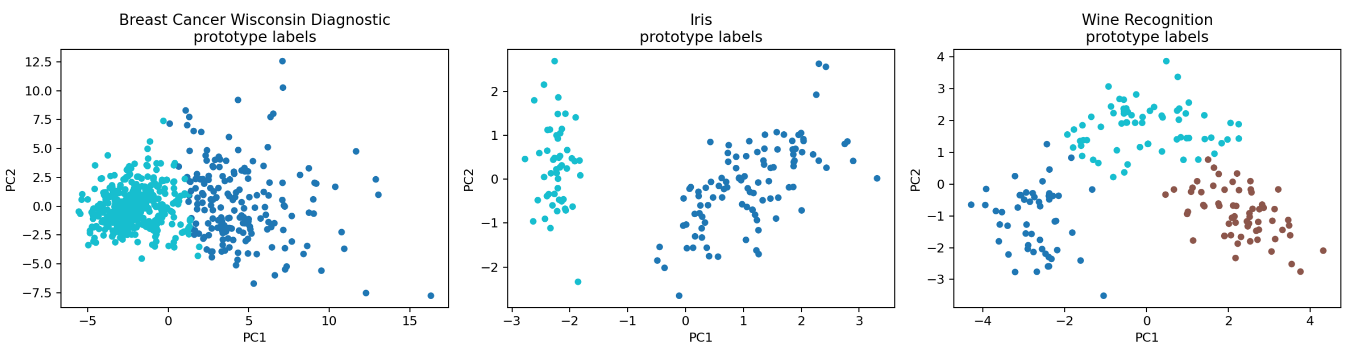

Figure 4: Best prototype-based clusters.

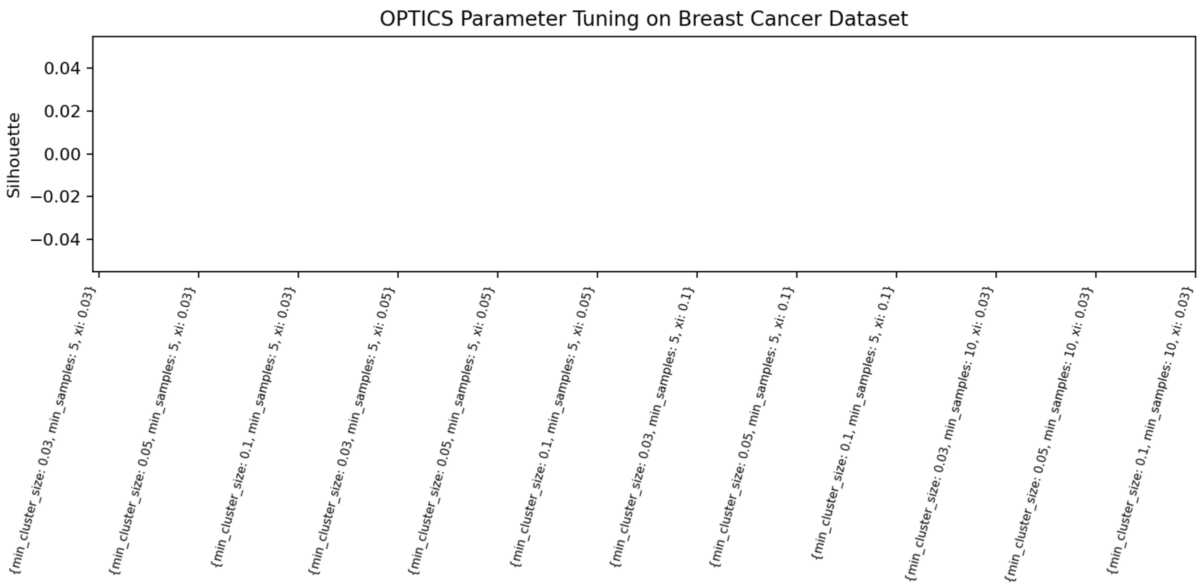

Figure 5: OPTICS parameter tuning on the density-selected dataset.

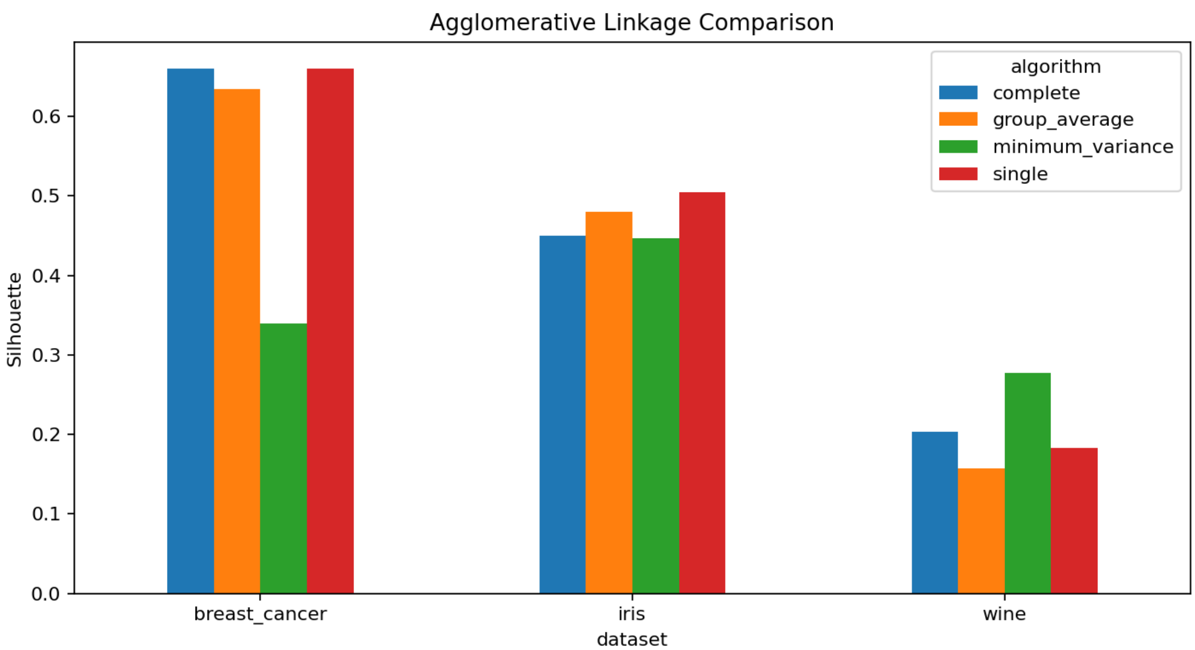

Figure 6: Agglomerative linkage comparison.

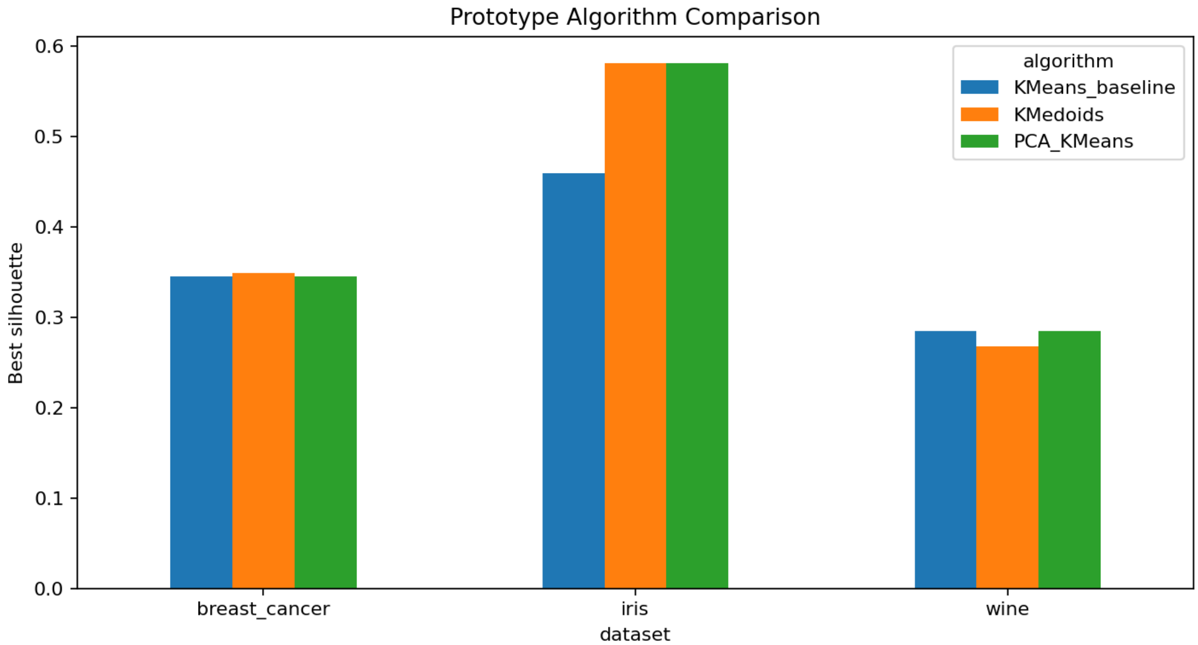

Figure 7: Prototype algorithm comparison.

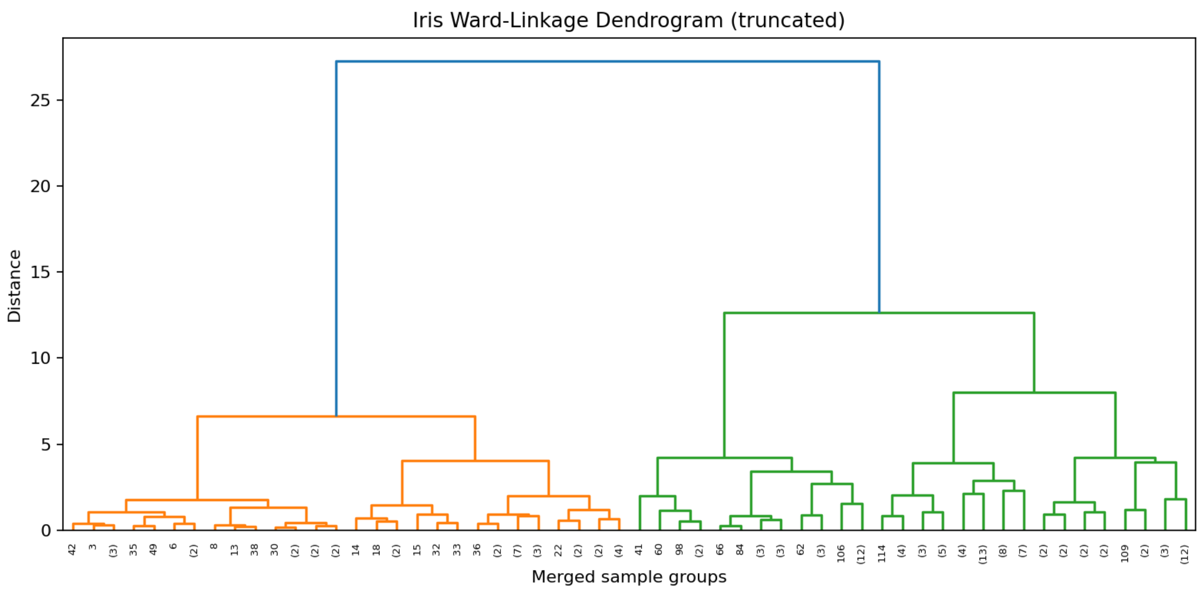

Figure 8: Truncated Iris dendrogram.

Inference

The Breast Cancer density hypothesis is only partly supported: OPTICS identifies dense substructure and noise, but standard tabular scaling still favors centroid-like separation for some metrics. The Iris hierarchical hypothesis is supported by Ward linkage and the dendrogram structure. The Wine prototype hypothesis is supported by strong KMeans/PCA+KMeans results relative to the alternatives.